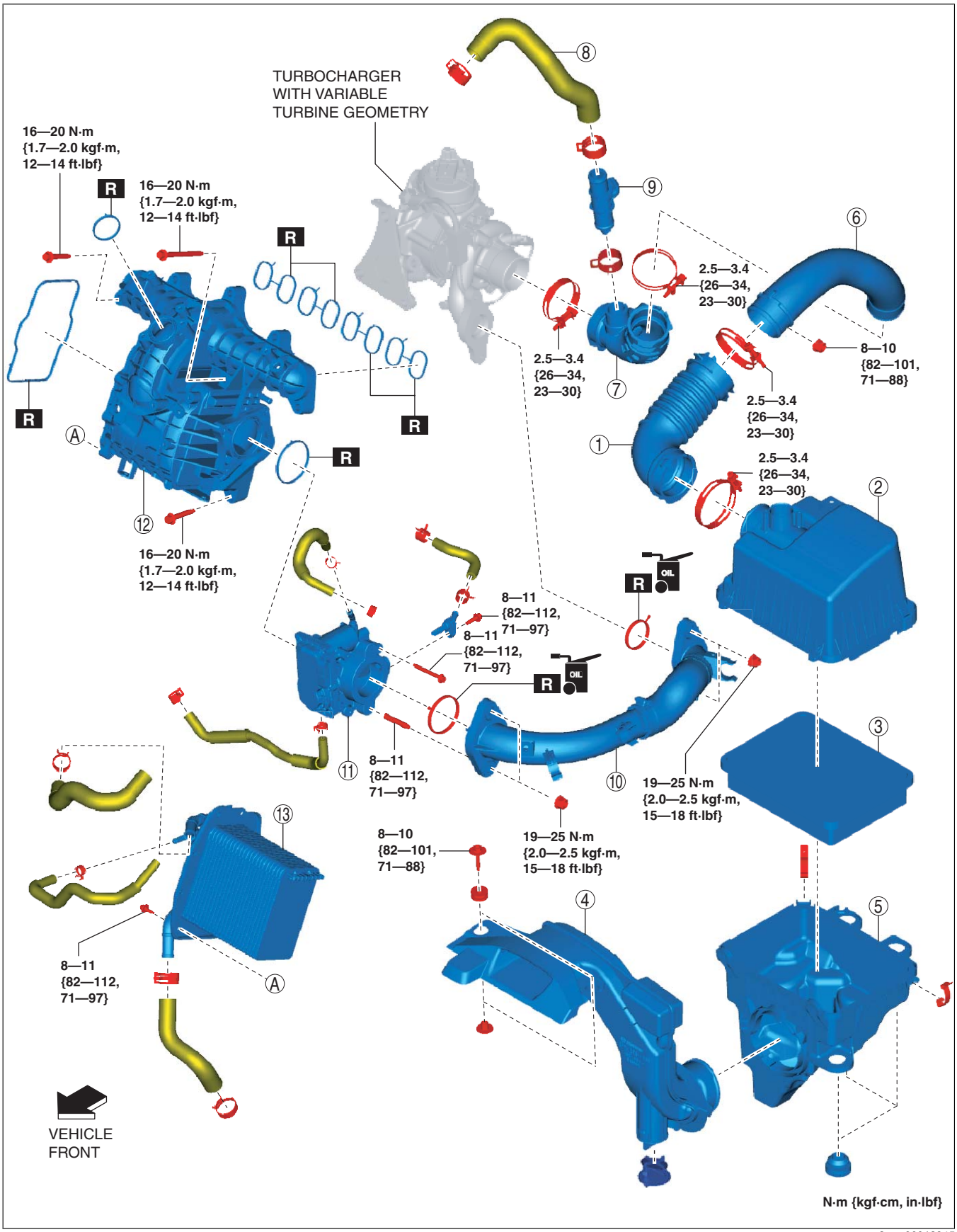
INTAKE-AIR SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

id0113q2801900

Warning

- **A hot engine and intake air system can cause severe burns. Turn off the engine and wait until they are cool before removing the intake-air system.**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove in the order shown in the figure.
3. Install in the reverse order of removal.



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1	Air hose (See Air Hose Installation Note.)
2	Air cleaner cover (See Air Cleaner Cover Removal Note.) (See Air Cleaner Cover Installation Note.)

3	Air cleaner element
4	Fresh-air duct (See Fresh-air Duct Removal Note.) (See Fresh-air Duct Installation Note.)

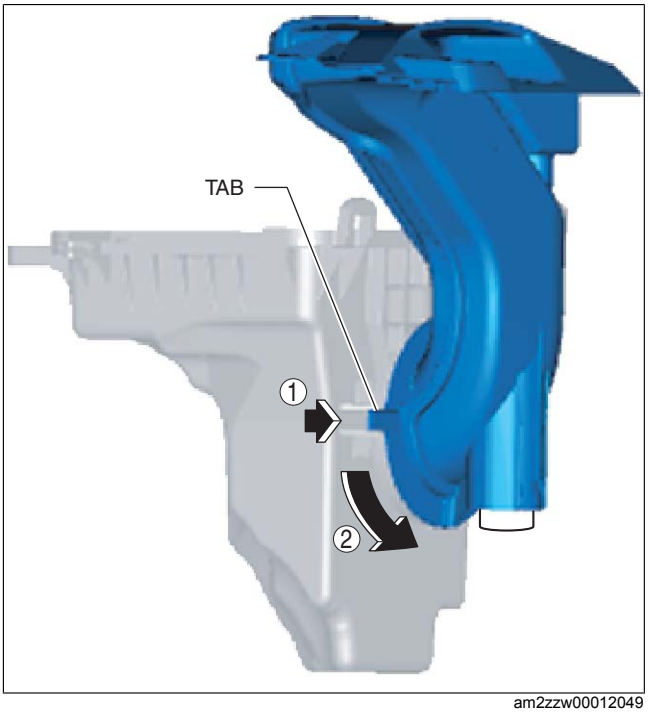
5	Air cleaner case (See Air Cleaner Case Installation Note.)
6	Air inlet pipe
7	Turbocharger air inlet hose (See Turbocharger Air Inlet Hose Removal Note.) (See Turbocharger Air Inlet Hose Installation Note.)
8	Breather hose (See Breather Hose Installation Note.)
9	Blow-by heater
10	Turbocharger air outlet pipe (See Turbocharger Air Outlet Pipe Removal Note.) (See Turbocharger Air Outlet Pipe Installation Note.)
11	Intake shutter valve (See INTAKE SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
12	Intake manifold (See Intake Manifold Removal Note.) (See Intake Manifold Installation Note.)
13	Water-cooled charge air cooler (See WATER-COOLED CHARGE AIR COOLER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Air Cleaner Cover Removal Note

1. Remove the MAF sensor/IAT sensor No.1. (See MASS AIR FLOW (MAF) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Fresh-air Duct Removal Note

1. Remove the following parts as a single unit:
   - Air cleaner case
   - Fresh-air duct
2. While moving the tabs in the direction shown in the figure, rotate the fresh air duct.
3. Remove the fresh-air duct.



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Turbocharger Air Inlet Hose Removal Note

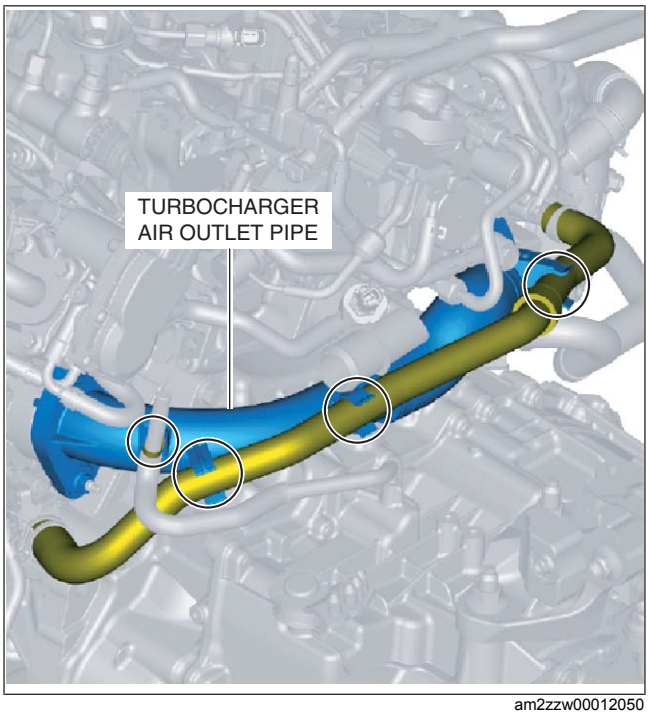
1. Remove the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Set the DC-DC converter (i-ELOOP) aside so that it does not interfere with the servicing. (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
3. Disconnect the blow-by heater connector.
4. Remove the following parts as a single unit:
   - Turbocharger air inlet hose
   - Blow-by heater
   - Breather hose
5. Remove the turbocharger air inlet hose.

Turbocharger Air Outlet Pipe Removal Note

Note

- The hose clamp installed to the turbocharger air outlet pipe is not removed or tightened.

1. Detach the hose and hose clip shown in the figure.
2. Remove the turbocharger air outlet pipe.



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Intake Manifold Removal Note

1. Remove the coolant reserve tank. (See COOLANT RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Remove the EGR pipe. (See EGR PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the water-cooled charge air cooler reserve tank. (See WATER-COOLED CHARGE AIR COOLER RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Disconnect the connectors of the following parts: (The sensor unit is not removed.)
   - MAP sensor/boost air temperature sensor (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR/ BOOST AIR TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - IAT sensor No.2 (See INTAKE AIR TEMPERATURE (IAT) SENSOR NO.2 REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Disconnect the wiring harness clip installed to the intake manifold.
6. Remove the intake manifold.

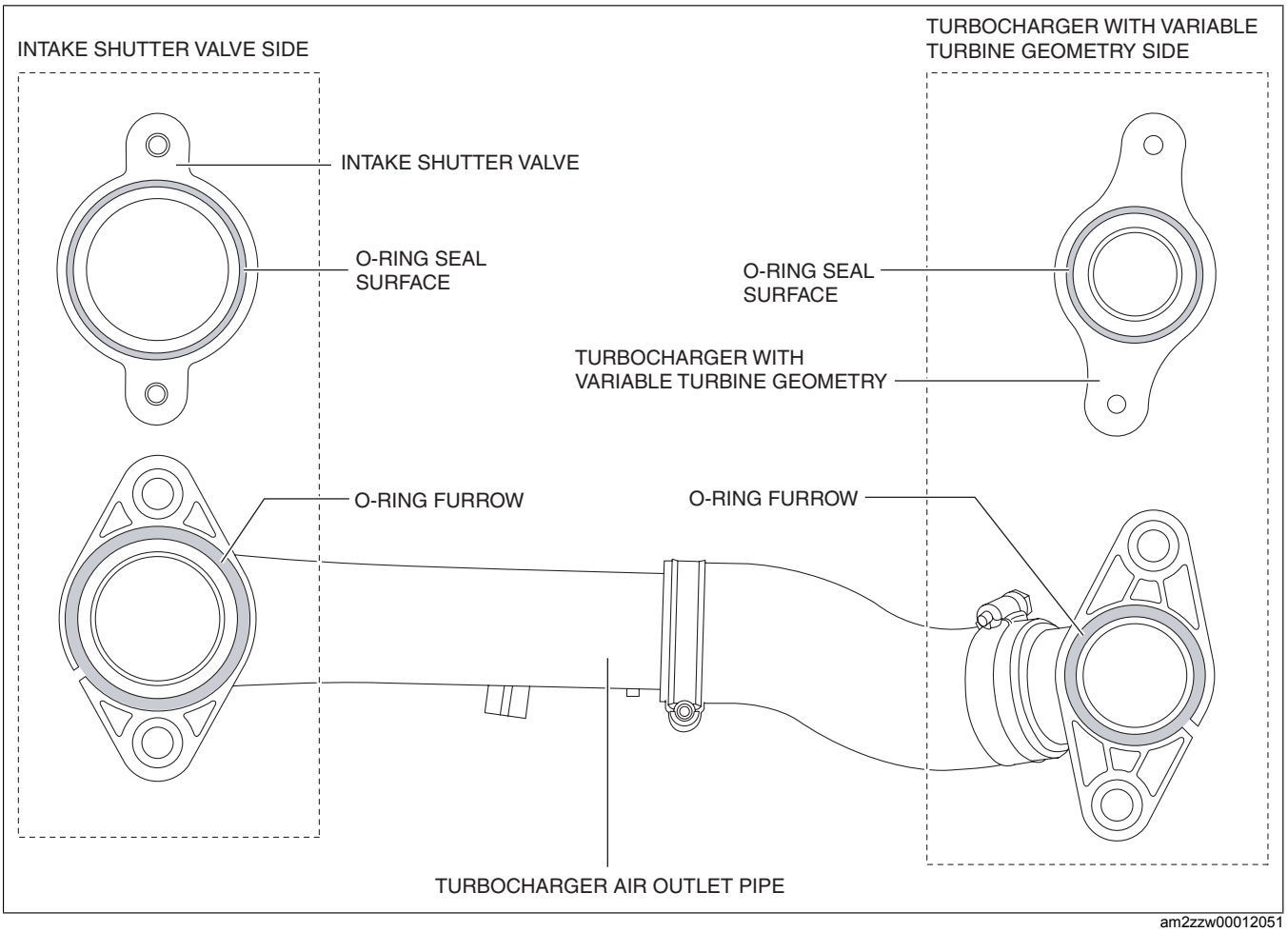
Intake Manifold Installation Note

1. Install the intake manifold.
2. Connect the wiring harness clip installed to the intake manifold.
3. Connect the connectors of the following parts:
   - MAP sensor/boost air temperature sensor (See MANIFOLD ABSOLUTE PRESSURE (MAP) SENSOR/ BOOST AIR TEMPERATURE SENSOR REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - IAT sensor No.2 (See INTAKE AIR TEMPERATURE (IAT) SENSOR NO.2 REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Install the water-cooled charge air cooler reserve tank. (See WATER-COOLED CHARGE AIR COOLER RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
5. Install the EGR pipe. (See EGR PIPE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Install the coolant reserve tank. (See COOLANT RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

Turbocharger Air Outlet Pipe Installation Note

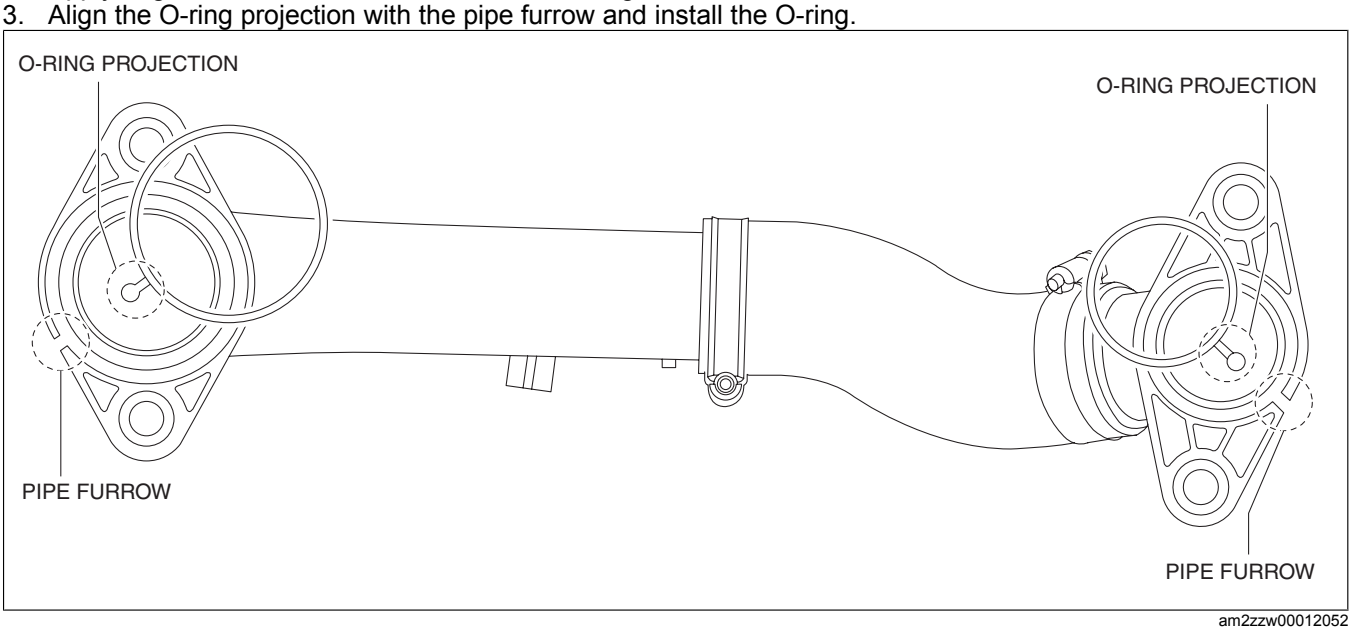
Note
- The hose clamp installed to the turbocharger air outlet pipe is not removed or tightened.

1. Remove foreign matter (sand or dust) on the seal surfaces of the variable geometry turbocharger and intake shutter valve O-ring, and the grooved area of the O-ring of the turbocharger air outlet pipe.



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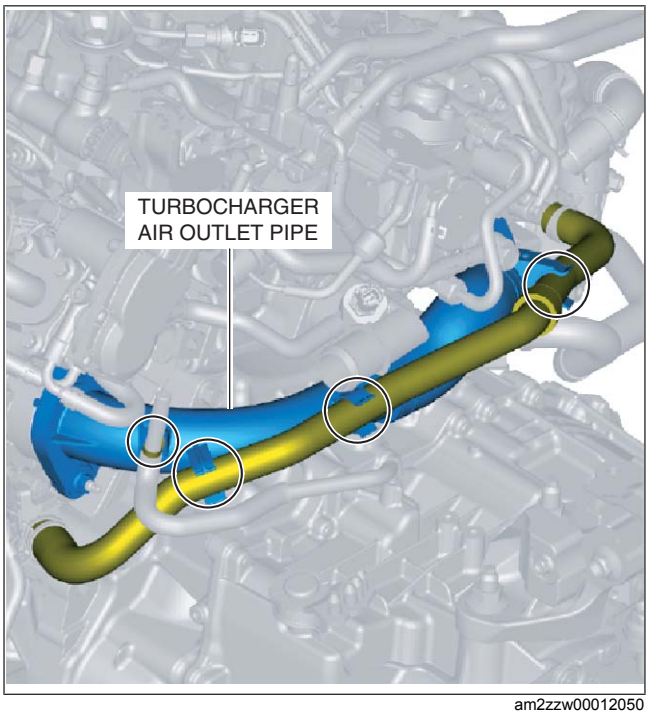
2. Apply engine oil to the entire surface of the O-ring.
3. Align the O-ring projection with the pipe furrow and install the O-ring.



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4. Install the turbocharger air outlet pipe.

5. Connect the hose and hose clip as shown in the figure.

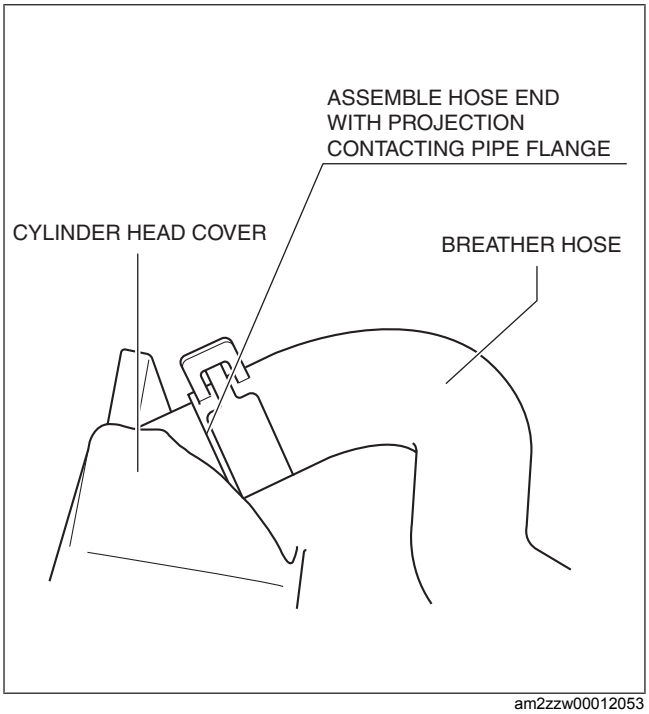


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Breather Hose Installation Note

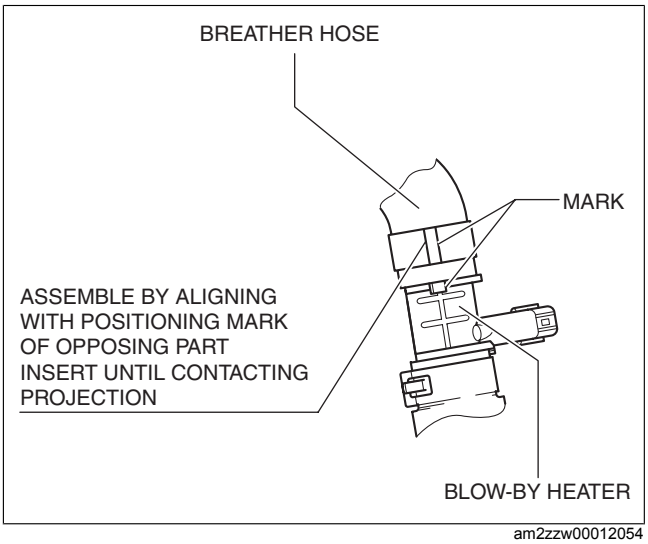
1. Install the breather hose as shown in the figure.

Cylinder head cover side



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Blow-by heater side

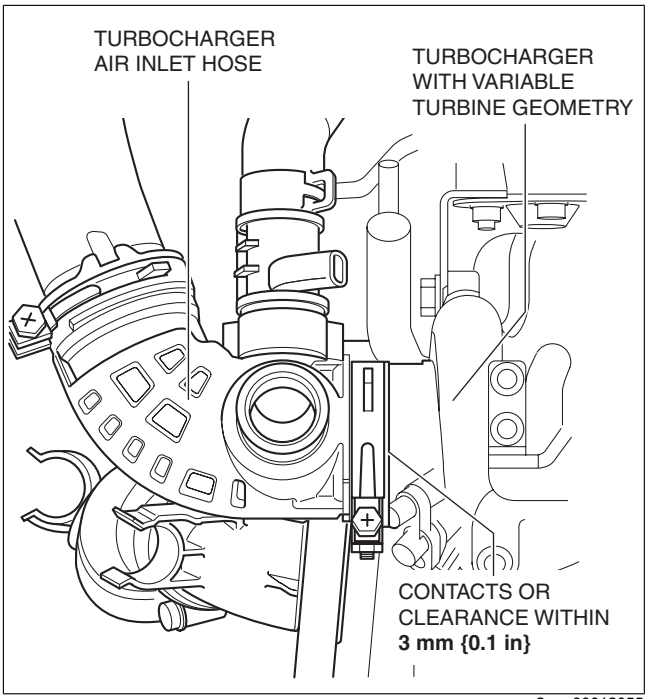


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Turbocharger Air Inlet Hose Installation Note

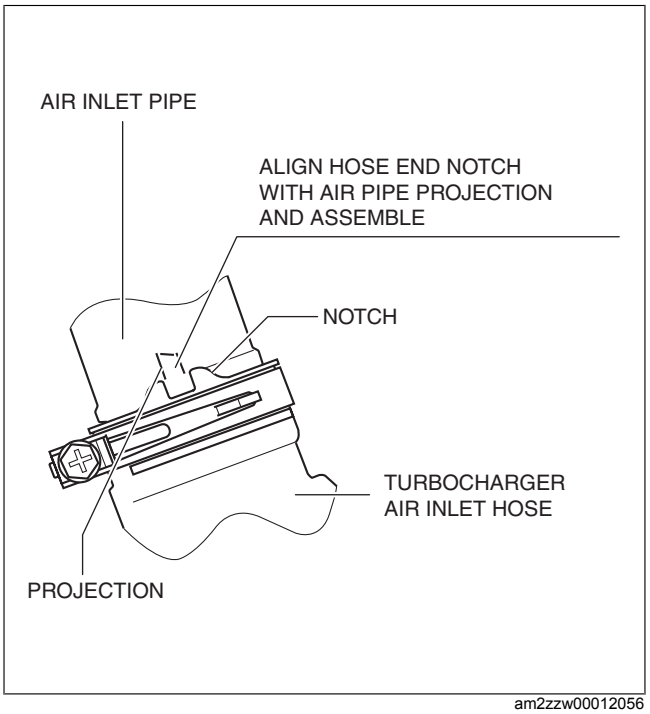
1. Install the turbocharger air inlet hose as shown in the figure.

Turbocharger with variable turbine geometry side



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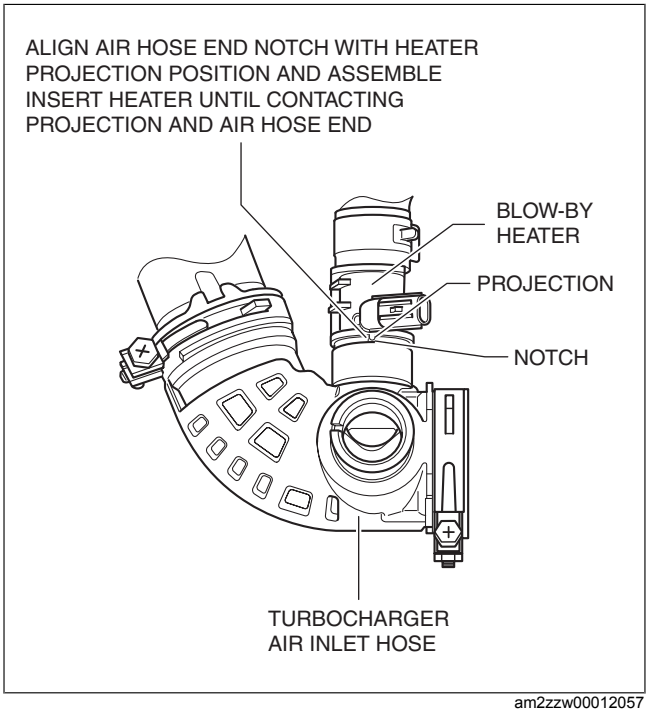
Air inlet pipe side



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Blow-by heater side

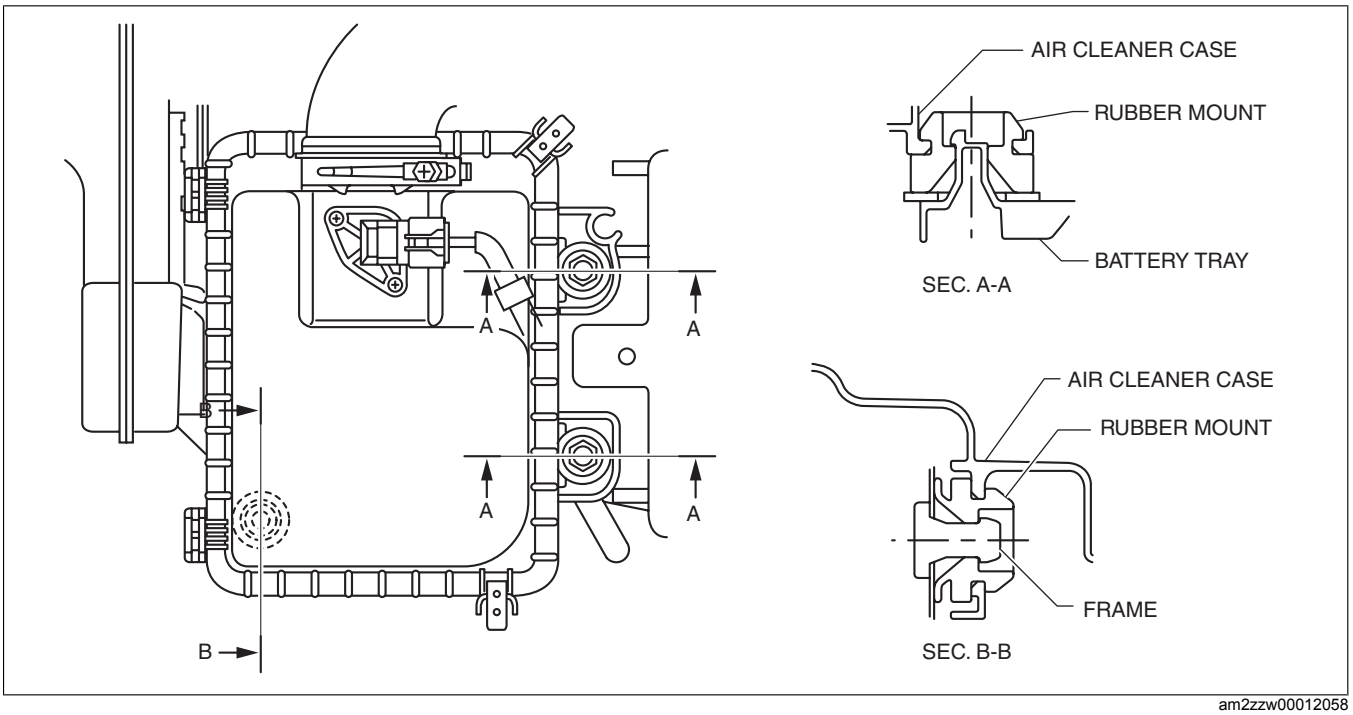
2. Install the following parts as a single unit.
   - Breather hose
   - Blow-by heater
   - Turbocharger air inlet hose
3. Connect the blow-by heater connector.
4. Install the DC-DC converter (i-ELOOP). (With i-ELOOP) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].)
5. Install the battery and the battery tray. (See BATTERY REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)



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Air Cleaner Case Installation Note

1. Install the air cleaner case as shown in the figure.



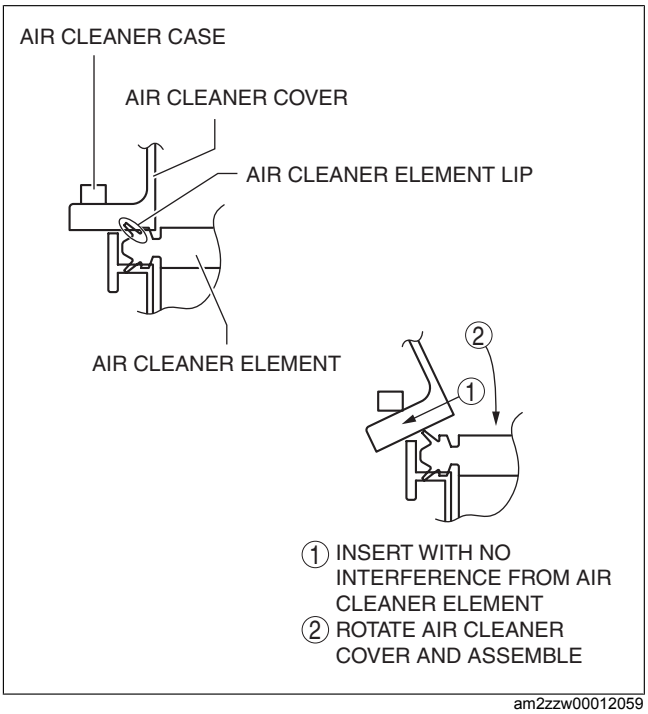
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Fresh-air Duct Installation Note

1. Install the fresh-air duct to the air cleaner case.
2. Install the fresh-air duct and air cleaner case as a single unit.

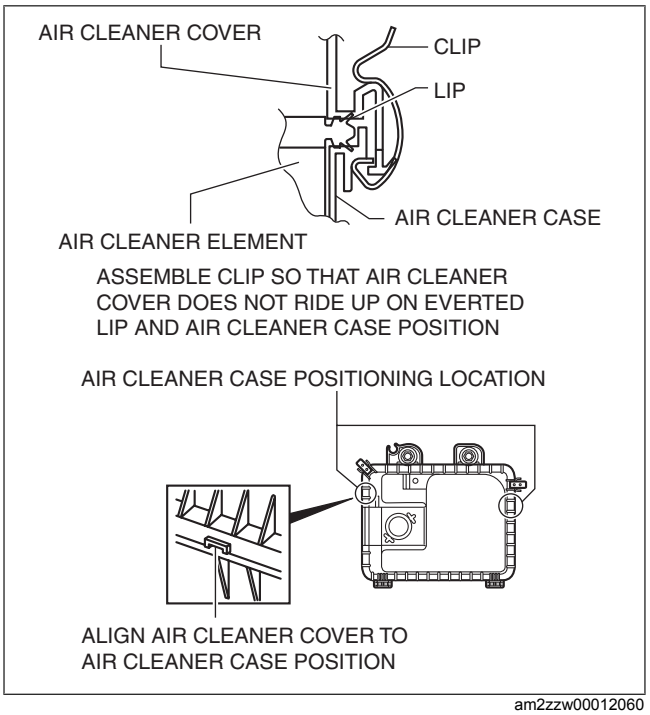
Air Cleaner Cover Installation Note

1. Install the air cleaner cover as shown in the figure.



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2. Install the air cleaner cover and air cleaner case for clip as shown in the figure.
3. Install the MAF sensor/IAT sensor No.1. (See MASS AIR FLOW (MAF) SENSOR/INTAKE AIR TEMPERATURE (IAT) SENSOR NO.1 REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)

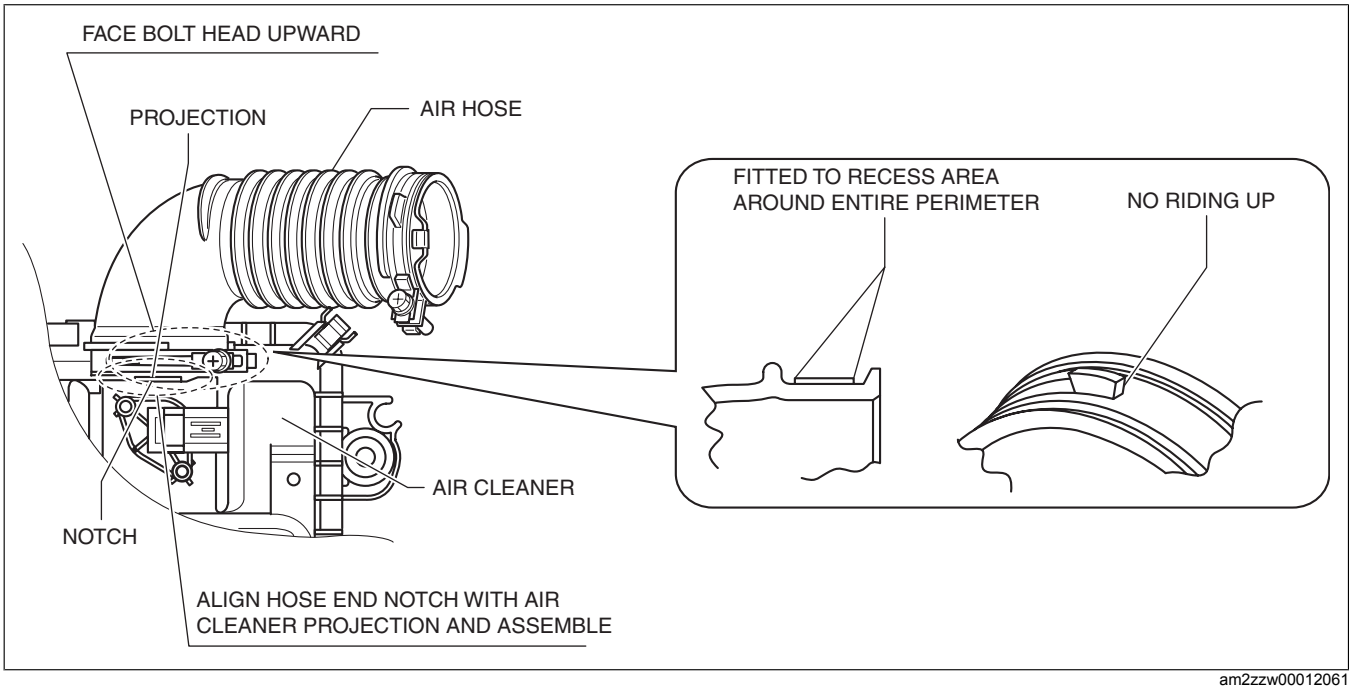


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Air Hose Installation Note

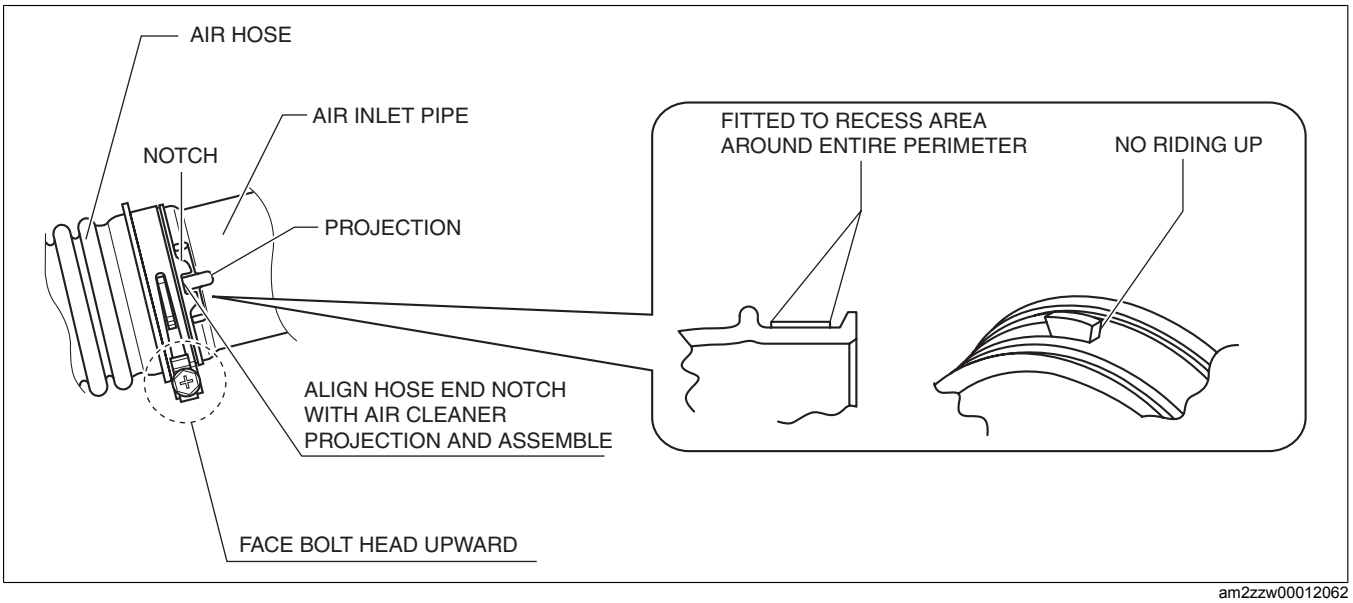
1. Install the air hose as shown in the figure.

Air cleaner side



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Air inlet pipe side



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